Playback Overview

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The Playback feature in GCXONE enables users to review recorded video footage from any integrated camera or device. It provides powerful tools for investigating incidents, reviewing events, and analyzing historical video data with synchronized multi-camera playback, event navigation, and advanced search capabilities.

Key Capabilities

Unified Timeline (Smart Timeline)

- Synchronized timeline: View recordings from multiple cameras on a single timeline with perfect time alignment
- Event Markers: Visual indicators showing when alarms, motion, or other events occurred
- Calendar Integration: Quick navigation to specific dates and times
- Recording Gaps: Visual indication of gaps in recording coverage
- Video Activity Search: Specialized search tool that displays all received events within preferred filtration criteria, making it easy to audit specific types of alarms or time periods

Playback Controls

- Standard controls: Play, pause, stop, rewind, and fast-forward
- Speed Control: Variable playback speeds from 1x up to 32x
- Frame-by-Frame: Step through individual frames for detailed analysis
- Jump to Time: Instantly navigate to specific timestamps
- Jump to Event: Quick navigation to event markers on timeline

Multi-Camera Playback

- Synchronized playback: View multiple cameras at the exact same point in time
- Grid Views: Monitor multiple camera feeds simultaneously
- Independent Controls: Control each camera playback independently if needed
- Comparison Mode: Compare views from different cameras side-by-side

Search and Navigation

- Time Search: Jump to specific date and time
- Event Search: Search for specific event types such as motion and alarms
- Calendar Navigation: Click on calendar dates to view recordings
- Bookmarks: Save specific timestamps for quick access

Video Export

- Export selected video segments in various formats
- Choose export quality and resolution
- Batch export multiple clips simultaneously
- Download clips or save to cloud storage

How It Works

Video is recorded on edge devices such as cameras and NVRs. GCXONE indexes recording metadata and event information, then generates a unified timeline from multiple device recordings. When a user requests playback, GCXONE retrieves the video from device storage and streams it to the user interface.

Recording Sources

- Edge Devices: Direct recording on cameras or NVRs
- VMS Platforms: Recordings stored in Milestone, Avigilon, and other VMS systems
- Cloud Storage: Recordings stored in cloud if configured
- Hybrid Storage: Combination of local and cloud storage

Use Cases

Incident Investigation

- Review video footage from incidents
- Analyze sequence of events across multiple cameras
- Export video clips as evidence with timestamps

Event Review

- Review alarm events to determine cause
- Analyze motion detection events
- Investigate false alarms and verify system performance

Forensic Analysis

- Detailed frame-by-frame analysis
- Multi-camera reconstruction of events
- Export clips for legal or compliance purposes

System Monitoring

- Review recording quality and coverage
- Identify recording gaps or issues
- Monitor storage usage and retention

Recording Requirements

For playback to work properly:

- Recording must be enabled on devices
- Sufficient storage space must be available on the device

- Devices must be synchronized with an NTP server
- Stable network connection between device and GCXONE is required
- Continuous or event-based recording schedule must be configured

Best Practices

Timeline Navigation

- Use event markers to quickly find relevant footage
- Use calendar for navigating to specific dates
- Bookmark important timestamps for quick access

Multi-Camera Review

- Use synchronized playback for event correlation
- Compare views from different angles
- Focus on relevant cameras for specific events

Video Export

- Export only necessary segments to save storage
- Include buffer time before and after events
- Organize exports with descriptive names

Performance Optimization

- Limit number of concurrent playback streams
- Close unused playback windows
- Use Local Mode for on-site workstations

Limitations and Considerations

Network Bandwidth

- Playback requires sufficient bandwidth
- Multiple concurrent streams increase bandwidth needs
- Consider Local Mode for high-bandwidth scenarios

Storage Availability

- Playback limited to available recordings
- Retention policies affect historical availability
- Recording gaps limit playback in those periods

Device Compatibility

- Playback features vary by device type
- Older devices may have restrictions
- VMS integration may have specific requirements